IMAGE PROCESSING TECHNIQUES – EXTENDED CLASS ACTIVITIES WORKSHEET

UG Level | Duration: 10 Minutes per Activity

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Activity 13: Mathematical Morphology

Fill the blanks:

Erosion shrinks objects.

Dilation expands objects.

Opening = Erosion + Dilation

Closing = Dilation + Erosion

How does morphology help in removing small noise?

Morphological operations help remove small noise by using a structuring element to modify the shapes in a binary image. The opening operation, which consists of erosion followed by dilation, removes small unwanted objects and isolated noise while preserving the main objects. Thus, morphology helps clean the image and improve its quality for further processing.